

On the Dissemination Latency of Cognitive Radio Networks Under General Node Mobility

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Abstract—Dissemination latency is critical to the applications of cognitive radio networks, which have become an important component of current communication infrastructure. This paper investigates the distribution of dissemination latency in a cognitive radio network where licensed users (primary users) are static and cognitive radio users (secondary users) are moving under general mobility, which provides fundamental understanding of the fastest information delivery that a mobile cognitive radio network is able to accommodate. We show that the dissemination latency depends on the stationary spatial distribution and mobility capability α (characterizing the region that a mobile secondary user can reach) of secondary users. Given any stationary spatial distribution, we find that there exists a critical value on α , below which the latency is heavy-tailed and above which the right tail of the distribution is bounded by some Gamma random variable. Moreover, we find that although the traffic and spatial density of primary users inversely impact the expected latency, they make no influence on the tail distribution of the latency. The results in this paper are validated through numerical studies and can advance our understanding of CR network performance.

I. INTRODUCTION

Today's spectrum is regulated by governmental agencies and is assigned to licensed users (also called *primary users*) on a long term basis in large geographical regions. A recent report from FCC reveals that under this static allocation, merely 5% ~ 15% of the spectrum is utilized on average [1]. As a promising solution to this spectrum under-utilization, cognitive radio networks have become an important component of current communication infrastructure. In cognitive radio networks, each secondary node is equipped with a *cognitive radio* and thus can access multiple channels without interfering with the licensed nodes. Therefore, cognitive radio networks facilitate the coexistence of disparate waveform technologies in the same network, which makes them suitable to a variety of application scenarios, such as military networks, emergency networks, cognitive mesh networks and leased networks [2].

Recently, there has been intensive research on understanding and optimizing the capacity and new functions, such as spectrum sensing, spectrum mobility and spectrum sharing, of CR networks [2]–[6]. While these research results have greatly advanced our understanding of architecture and the capacity of CR networks, another equally important performance indicator, *dissemination latency*, has received less attention. In the QoS-sensitive applications, the latency is more QoS relevant than

other performance metrics, such as the total network capacity. For example, when a CR network is used for emergency rescue in the aftermath of traffic accidents (e.g. vehicular networks), we must ensure that *help* or *warning* message can be disseminated to a chosen destination within a predetermined time. This motivates *dissemination latency* study in this paper.

Latency has been studied for multihop wireless networks in pioneering work [7]–[10]. The work in [7], [8] studied the dissemination latency in *fully connected* multi-hop wireless networks and showed that there exist bounds on the latency and these bounds are tight when the number of nodes are large enough. The work in [9], [10] further showed that the asymptotic latency scales at least linearly with the transmission distance in the *percolated* wireless networks. Although these results have greatly advanced our understanding of the nature of latency, a few interesting questions remain unanswered. First, wireless nodes in these works are assumed to be static. However, in many cognitive communication applications (e.g. cognitive vehicular networks or cognitive military networks), secondary users usually need to move around to obtain “better channel opportunity” or “more security”. What is the impact of mobility on latency? Moreover, the existing works only derive the asymptotic latency which assumes that the number of wireless nodes is infinite or approaches to infinity. However in reality, the number of nodes is finite. Then what is the dissemination latency for a network with finite users? We attempt to provide answers to these questions in this paper.

Specifically, we consider a mobile cognitive radio network where primary users are static and randomly distributed, and the secondary users move around with arbitrary mobility patterns. We first provide a general mobility model which captures most of the features of the existing mobility models and takes *spatial inhomogeneities* into account. We show that dissemination latency depends on the spatial distribution and mobility capability α (characterizing the region that a mobile user can reach) of secondary users. We find that given any spatial distribution of secondary users, there exists a critical value on α , below which the latency follows *heavy-tailed* distribution and above which the right tail of its distribution is bounded by some Gamma random variable. Furthermore, we find that although the traffic and spatial density of primary users inversely impact the expected latency, they have no influence on the “distribution” of the latency in finite mobile cognitive radio networks.

This paper is organized as follows. In Section II, we es-

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establishes the network and mobility models, and defines the concepts and notations to be used later on. We derive the distribution of dissemination latency in a finite cognitive radio networks under general mobility in Section III. In Section IV, we use simulations to validate our analysis, followed by the conclusions in Section V.

II. SYSTEM ASSUMPTIONS AND NOTATION

In this section, we first describe the mobility models used in this paper and then define *dissemination latency* formally.

A. Mobility Models

We consider a family of mobile cognitive radio networks which is denoted by $\mathcal{M}(\Phi, \Psi, \alpha)$ and is characterized by three parameters Φ, Ψ, α . For a given choice of parameters, we build a sequence of networks $\mathcal{F}_{m,n}$ indexed by the number of primary users m and the number of secondary users n . Secondary users $\{v_i\}_{i=1}^n$ are supposed to move over a bi-dimensional Torus surface $\mathcal{D}_n$ of size $L_n \times L_n$ (to avoid border effects) and simultaneously, m static primary users $\{u_k\}_{k=1}^m$ are randomly distributed in $\mathcal{D}_n$. We assume that L_n scales with n as n^δ , with $0 \leq \delta \leq 1/2$. The extreme cases are: $\delta = 0$, for which the network area remains constant and node density increases linearly with n ; $\delta = 1/2$, for which the network area increases linearly with n and node density remains constant. δ determines how the network area behaves as the number of nodes increases. And in this paper, we focus on the scenario that $L_n = \sqrt{\frac{n}{\lambda}}$ ($\delta = 1/2$). However, our results can be easily extended to the cases with different L_n .

Anisotropic mobility is taken into account in our model. Let $v_i(t)$ denote the location of the secondary user v_i at time t (since primary users are static, u_k also denotes the location of primary user u_k for simplicity). Let $V(t) = (v_1(t), \dots, v_n(t))$ and $U(t) = (u_1, \dots, u_m)$ denote the vector of locations, and $V(0)$ and $U(0)$ be the initial distribution, of secondary users and primary users respectively. We consider that the stationary spatial distribution of a secondary user is generally non-uniform over the space. A secondary user typically spends most of the time in a small region, and rarely visits the areas far away from it. We model this behavior by assuming that each secondary user v_i has a *home point*, located at v_i^h . Secondary users move “around” their home points according to independent stationary and ergodic processes.

To account for *spatial inhomogeneities* of the overall density of secondary users over $\mathcal{D}_n$, we assume that each home point v_i^h is associated to a fixed point v_i^c , which is called the *center point* of v_i . The center points $\{v_1^c, \dots, v_n^c\}$ are placed regularly at positions $(\frac{1}{2\sqrt{\lambda}} + \frac{i}{\sqrt{\lambda}}, \frac{1}{2\sqrt{\lambda}} + \frac{j}{\sqrt{\lambda}})$ with $0 \leq i \leq \sqrt{n} - 1$ and $0 \leq j \leq \sqrt{n} - 1$ (see Fig. 1). We describe the distribution of the home point v_i^h by a non-increasing probability density function $\Phi_i(X) = \Phi(X - v_i^c)$, which is assumed to be invariant in all directions. Moreover, we describe the probability density of secondary user v_i around v_i^h by a non-increasing and direction-invariant function $\Psi_i(x) = \Psi(x - v_i^h)$. We assume that Ψ_i is non-zero in and only in a region characterized by a constant α ; that is, $\Psi_i(x) = \Psi(x - v_i^h) > 0$ when $\|x - v_i^h\| < \alpha$ and

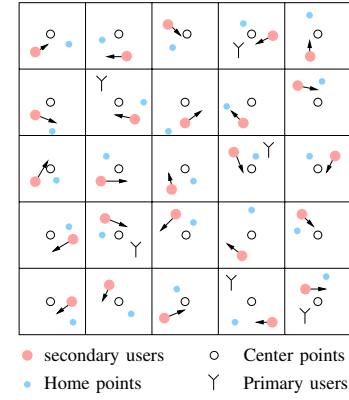


Fig. 1. A cognitive radio network under general mobility.

$\Psi_i(x) = \Psi(x - v_i^h) = 0$ otherwise. α is called *mobility capability* since 2α represents the *moving diameter* of secondary users. For simplicity, we assume that the locations of home points, and the mobility of all secondary users are characterized by the common functions $\Phi(x)$ and $\Psi(x)$, respectively. However, our analysis can be easily extended to the case in which home points follow different distribution functions $\Phi_i(x)$ and secondary users with different mobility patterns $\Psi_i(x)$ coexist. Fig. 1 describes an example of a cognitive radio network under general mobility.

Discussion: The class of mobile networks that we study are very general such that they account for a wide range of possible scenarios of realistic mobility processes. Notice that, as a special case, we can obtain the *Clustered Grid model* (see [11]) when the home point density functions $\Phi_i(x) = \delta(x - v_i^c)$, $\delta(x)$ being the Dirac impulse function. When $\Phi(x)$ is a constant function independent of x , we obtain *Uniform Anisotropic model* [11]. The case of static nodes uniformly deployed over the area (Gupta-Kumar case [12]) can be obtained by setting $\Phi(x)$ as a constant function independent of x and $\Psi_i(x) = \delta(x - v_i^h)$. And the *i.i.d* mobility model in [13] corresponds to the case when $\Psi(x)$ is a constant function independent of x and $\alpha = \infty$; furthermore, when $\alpha < \infty$, we obtain the *constrained i.i.d* model used in [10].

The general mobility can be characterized into different classes based on the distributions of the home points of secondary users. And for simplicity, we only focus on dissemination latency of cognitive radio networks under the following two class of mobility. And techniques used and results derived in this paper can be extended to the scenarios under any mobility.

- **Fixed Home Point (FHP)** mobility $\mathcal{M}(\Phi_F, \Psi, \alpha)$: Home points $\{v_i^h\}_{i=1}^n$ are fixed. Here $\Phi_F(x) = \delta(x - v_i^c)$ and this model is also called *Clustered Grid model* in [11].
- **Partially Random Home Point (PRHP)** mobility $\mathcal{M}(\Phi_P, \Psi, \alpha)$: The home point v_i^h is independently distributed in $\mathcal{O}_i$, where $\{\mathcal{O}_i\}_{i=1}^n$ denote the small regions in $\mathcal{D}_n$ partitioned by center points $\{v_i^c\}_{i=1}^n$, as shown in Fig. 1.

B. Interference Model

We assume that interference among simultaneous transmissions is described by the *protocol model* [12]. Particularly, denote R_I as the interference range of the primary users and r as the transmission range of secondary users. Secondary user v_i is allowed to transmit to v_j at time t , only if: i) there is no primary users in the neighborhood, i.e., $\|v_i(t) - u_k\| > R_I$ and $\|v_j(t) - u_k\| > R_I$ for any k ; and ii) $\|v_i(t) - v_j(t)\| < r$, where $\|\cdot\|$ denotes the Euclidean distance.

C. Notations and Metrics

By assuming that every node communicates with a randomly chosen destination, and serves as a relay for all the other S-D pairs, the trade-offs between throughput and delay in wireless networks under different mobilities has been extensively studied in related works [13], [14]. Instead of capacity-delay trade-offs, our target in this paper is to answer the question of how fast a message can be disseminated in mobile cognitive radio networks. Particularly, there is a *single message* to be delivered by a source secondary user v_s to a destination v_d . Intermediary secondary users can be used as relay nodes. The goal is to determine the distribution of the dissemination latency.

In a mobile environment, two nodes can exchange information and thus share a link if they can decode each other's signals at *some time*. Since different pairs of nodes can share links at different points in time, connectivity varies over time. Without considering the propagation delay, when the source v_s sends a message at time t_0 , all the nodes in the connected component of v_s receive the message instantaneously. The destination v_d may not receive this message if it is not in this connected component. As time goes on, nodes move, and the message is passed from message-carrying nodes to other nodes whenever they share communication links. Thus, the information is disseminated through the network and v_d may receive this message at some point as this process continues. The dissemination latency $\mathcal{T}_d$ can be defined as follows:

Definition 1: Denote $\mathbb{L}(t)$ as the set of communication links at time t . Given the initial node distribution $V(0)$ and $U(0)$, and mobility $\mathcal{M}(\Phi, \Psi, \alpha)$, let the *first hitting time* of secondary users v_i and v_j be

$$T_h(v_i, v_j) \triangleq \inf\{t \geq 0 : l_{ij} \in \mathbb{L}(t)\}$$

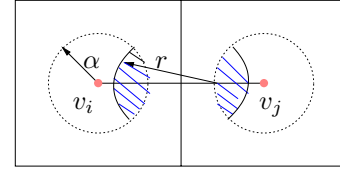
where l_{ij} denotes a communication link between v_i and v_j .

Let $\mathcal{A}(t)$ be the set of secondary users that have received the message sent at time 0 by time t . Define $\mathcal{T}_d$ as:

$$\mathcal{T}_d \triangleq \inf\{t \geq 0 : v_d \in \mathcal{A}(t)\}.$$

Then, $\mathcal{T}_d$ is the *latency* of information dissemination from source v_s to the randomly chosen destination v_d .

And for convenience, we further denote $d^{(t)}(v_i, v_j)$, $d_h(v_i, v_j)$ and $d_c(v_i, v_j)$ as the “distance” between secondary users v_i and v_j at time t , the “distance” between v_i^h and v_j^h , and the “distance” between v_i^c and v_j^c , respectively. Here the “distance” could be any p -norm distance function.



$$d_h(v_i, v_j) = \sqrt{\frac{1}{\lambda}}$$

Fig. 2. Calculation of the expected first hitting time under $\mathcal{M}(\Phi_F, \Psi, \alpha)$.

III. THE DISTRIBUTION OF DISSEMINATION LATENCY $\mathcal{T}_d$

In this section we are interested in the distribution of dissemination latency $\mathcal{T}_d$ in finite mobile cognitive radio networks, which clearly depends on mobility $\mathcal{M}(\Phi, \Psi, \alpha)$. In Section II-A, we mentioned that $\mathcal{M}(\Phi, \Psi, \alpha)$ can be classified into different classes based on the distribution of home points and we focus on two type of mobility: **FHP** mobility and **PRHP** mobility. Particularly, in this section, we will study the distribution of $\mathcal{T}_d$ under **FHP** mobility and **PRHP** mobility. Results derived in this section can be easily extended to scenarios with more general mobility.

A. Latency Distribution under FHP Mobility $\mathcal{M}(\Phi_F, \Psi, \alpha)$

Note that two secondary users v_i and v_j may communicate with each other directly only when $2\alpha + r > d_h(v_i, v_j)$. Because in FHP mobility, home points are fixed and form a regular square grid, with the distance between neighboring home points being $\sqrt{\frac{1}{\lambda}}$ (see Fig. 1), therefore, the number of secondary users that can communicate directly with v_i under FHP mobility is finite. This implies that the source v_s may not communicate directly with v_d . To deliver information to v_d , v_s needs other secondary users as relay. To investigate the dissemination latency of this process, we will identify the transmission time through a path joining v_s and v_d , which provides an upper bound on $\mathcal{T}_d$. In the following lemma, we first study the first hitting time between two neighboring secondary users v_i and v_j $T_h(v_i, v_j)$. Particularly, we mathematically prove that $T_h(v_i, v_j)$ decays at least exponentially fast if the mobility capability α is larger than $\frac{\sqrt{\frac{1}{\lambda}} - r}{2}$.

Lemma 1: Given initial node distribution $V(0)$ and $U(0)$, and mobility $\mathcal{M}(\Phi_F, \Psi, \alpha)$, for two secondary users v_i and v_j with $d_c(v_i, v_j) = \sqrt{\frac{1}{\lambda}}$, we have $T_h(v_i, v_j) = \infty$ if $\alpha < \frac{\sqrt{\frac{1}{\lambda}} - r}{2}$; otherwise, $E(T_h(v_i, v_j)) < \infty$ and $\mathbb{P}(T_h(v_i, v_j) > t) \leq e^{-c_1 t}$ for sufficiently large t . c_1 is some positive constant.

Proof: Denote by $\mathcal{B}(x, a)$ the circular region centered at x with radius a . Note that any two nodes v_i and v_j can communicate with each other directly at time t if and only if $d^{(t)}(v_i, v_j) \leq r$. Since v_i and v_j can appear only within $\mathcal{B}(v_i^h, \alpha)$ and $\mathcal{B}(v_j^h, \alpha)$ respectively, thus if $2\alpha + r < d_h(v_i, v_j)$, then $d^{(t)}(v_i, v_j) > r$ for all $t \geq 0$. Hence, $T_h(v_i, v_j) = \infty$ if $\alpha < \frac{d_h(v_i, v_j) - r}{2}$. Now assume that $2\alpha + r > d_h(v_i, v_j)$. Denote $\mathcal{E}_t$ as the event that there exists no communication link between v_i and v_j at any time point t and denote $\varepsilon = \mathbb{P}(\mathcal{E}_t)$. As shown in Fig. 2, in this scenario, when v_i and v_j are in the shaded

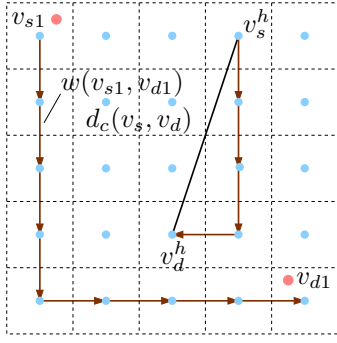


Fig. 3. Examples of Manhattan paths between sources and destinations.

region, they may communicate with each other directly. By categorizing the scenario into cases whether the shaded regions intersect with each other and whether primary users appear in the neighborhood, we can calculate the probability ε by using *conditional probability* techniques. The probability ε depends on the stationary distribution of secondary users Ψ and the traffic and density of primary users. Since Ψ_i is non-zero at any point in $\mathcal{B}(v_i^h, \alpha)$, we have $0 < \varepsilon < 1$ by assuming the density and traffic of primary users are low.

Denote ρ as a *renewal* interval for secondary user v_i , i.e., for any $t > 0$, $\{v_i(t') : t' \leq t\}$ and $\{v_i(t'' + \rho) : t'' \geq t\}$ are independent; i.e., after ρ , v_i completely forgets where it was ρ ago. ρ depends on the specific mobility model. Observe that for the first hitting time $T_h(v_i, v_j)$, we have $\{T_h(v_i, v_j) > t\} \subset \cap_{s \in \mathcal{I}_t} \mathcal{E}_s$ for any subset $\mathcal{I}_t \subset (0, t]$. And if we choose $\mathcal{I}_t = \{\rho, 2\rho, \dots, \lfloor \frac{t}{\rho} \rfloor \rho\}$, it follows that $\mathcal{E}_{t_i}$ are all independent for any $t_i \in \mathcal{I}_t$ since ρ is the *renewal* interval for v_i and v_j .

$$\mathbb{P}(T_h(v_i, v_j) > t) \leq \mathbb{P}(\cap_{s \in \mathcal{I}_t} \mathcal{E}_s) = \prod_{s \in \mathcal{I}_t} \mathbb{P}(\mathcal{E}_s) = \varepsilon^{\lfloor \frac{t}{\rho} \rfloor} \sim e^{-c_1 t},$$

where $c_1 = -\log \varepsilon / \rho$. This completes the proof. ■

Lemma 1 implies that two neighboring secondary users can communicate with each other within finite time with high probability only when $\alpha > \frac{\sqrt{\frac{1}{\lambda}} - r}{2}$ and we are only interested in $\mathcal{T}_d$ in this case. Before proceeding, we introduce an important concept *coupling method*, which will be used throughout this paper.

Definition 2: If Z and Z' are random variables such that $\mathbb{P}(Z > z) \leq \mathbb{P}(Z' > z)$, we say that Z is *stochastically dominated* by Z' and write $Z \stackrel{\mathcal{D}}{<} Z'$; and if $Z \stackrel{\mathcal{D}}{<} Z'$, there exists a random variable $\hat{Z}'$ which has the same distribution of Z' such that $Z \leq \hat{Z}'$ ($\hat{Z}'$ is called a *coupling* of Z' [15].)

We now present the main result on $\mathcal{T}_d$ and $\mathcal{S}_d$ under FHP mobility.

Theorem 2: Given $V(0)$, $U(0)$ and $\mathcal{M}(\Phi_F, \Psi, \alpha)$, if $\alpha > \frac{\sqrt{\frac{1}{\lambda}} - r}{2}$, the dissemination latency $\mathcal{T}_d$ is *stochastically dominated* by a *Gamma distribution* $\text{Gamma}(2\sqrt{n}, c_1)$.

Proof: If $\alpha > \frac{\sqrt{\frac{1}{\lambda}} - r}{2}$, for any v_s and v_d , we can identify a path through which v_s first disseminates the message vertically until the messages reaches the secondary user whose home

point has the same horizontal coordinate with v_d^h , and then disseminates the message to v_d horizontally. This path is called a *Manhattan path* and two example of such paths are shown in Fig. 3. Note that the longest *Manhattan path* has at most $2\sqrt{n}$ hops (see $w(v_{s1}, v_{d1})$ in Fig. 3). Denote $\{Y_k\}_{k=1}^{\infty}$ as a sequence of independent random variables with the same distribution as $T_h(v_i, v_j)$. Thus, $\mathcal{T}_d \leq \sum_{k=1}^{2\sqrt{n}} Y_k$. By Lemma 1 and *coupling* method, we can find a sequence of independent random variables $\{\hat{Y}_k\}_{k=1}^{\infty}$ with exponential distribution $\text{exponential}(c_1)$ such that $Y_k < \hat{Y}_k$ for all k . Thus $\mathbb{P}(\mathcal{T}_d > t) \leq \mathbb{P}(\sum_{k=1}^{2\sqrt{n}} Y_k > t) \leq \mathbb{P}(\sum_{k=1}^{2\sqrt{n}} \hat{Y}_k > t) = \mathbb{P}(Y > t)$, where $Y = \sum_{k=1}^{2\sqrt{n}} \hat{Y}_k$. By *Moment generating function* technique ([16]), we know that Y follows a *Gamma distribution* $\text{Gamma}(2\sqrt{n}, c_1)$. This completes the proof. ■

Remark 1: From [16], we know that Gamma distribution is light-tailed distribution, i.e., it is *exponentially* bounded. Thus, Theorem 2 provides the condition when $\mathcal{T}_d$ has a light tail under FHP mobility.

B. Latency Distribution under PRHP Mobility $\mathcal{M}(\Phi_P, \Psi, \alpha)$

In the PRHP mobility, the home point v_i^h is randomly distributed in a small region $\mathcal{O}_i$ centering at v_i^c . And for two home points v_i^h and v_j^h in the regions $\mathcal{O}_i$ and $\mathcal{O}_j$ with $d_c(v_i, v_j) = \sqrt{\frac{1}{\lambda}}$ (called *neighboring regions*), $d_h(v_i, v_j)$ is a random variable with $\mathbb{P}(d_h(v_i, v_j) \leq \sqrt{\frac{5}{\lambda}}) = 1$. Considering that $\{\mathcal{O}_i\}_{i=1}^n$ are non-intersected and v_i can communicate with v_j directly only when $2\alpha + r > d_h(v_i, v_j)$, each v_i can communicate directly with only finite secondary users. This indicates that similar to the scenario with FHP mobility, the information dissemination between v_s and v_d under PRHP mobility probably needs other secondary users as relay. Similar to the previous subsection, we first rigorously prove that the first hitting between secondary users with home points in the neighboring regions is exponentially bounded.

Lemma 3: Given $V(0)$, $U(0)$, $\mathcal{M}(\Phi_P, \Psi, \alpha)$ and secondary users v_i and v_j with $d_c(v_i, v_j) = \sqrt{\frac{1}{\lambda}}$, if $\alpha < \frac{\sqrt{\frac{5}{\lambda}} - r}{2}$, $T_h(v_i, v_j)$ follows a *heavy tailed* distribution and $E(T_h(v_i, v_j)) = \infty$; otherwise, $\mathbb{P}(T_h(v_i, v_j) > t) \leq e^{-c_2 t}$ for sufficiently large t . c_2 is some positive constant.

Proof: See proof of Lemma 1. ■

Note that except the *mobility capability* α required to disseminate information to the whole network, the dissemination process from v_s to the destination v_d under PRHP mobility is similar to that under FHP mobility. Thus by similar proof to Theorem 2, we have the following results on $\mathcal{T}_d$.

Theorem 4: Given $V(0)$, $U(0)$ and mobility $\mathcal{M}(\Phi_P, \Psi, \alpha)$, if $\alpha > \frac{\sqrt{\frac{5}{\lambda}} - r}{2}$, the dissemination latency $\mathcal{T}_d$ is *stochastically dominated* by a *Gamma distribution* $\text{Gamma}(2\sqrt{n}, c_2)$.

C. Discussion

In this section, we have studied dissemination latency $\mathcal{T}_d$ in finite cognitive radio networks under general mobility models. We show that $\mathcal{T}_d$ depends on the spatial distribution and the

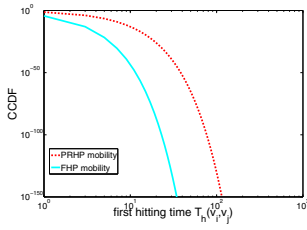


Fig. 4. CCDF of the first hitting time.

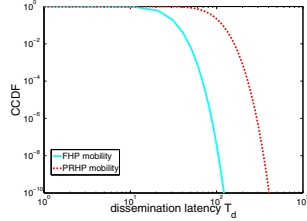


Fig. 5. CCDF of the dissemination latency.

mobility capability α of secondary users. Given any spatial distribution, we mathematically prove the critical value on α , crossing over which T_d turns from *heavy-tailed* distribution to *Gamma* distribution. Dissemination latency is one of the key metrics in wireless networks and central to routing algorithms. Our results thus provide guidelines on the design of mobility models with *Gamma* (light-tailed) dissemination latency, new protocols as well as their performance analysis. Furthermore, we must emphasize that although our results seem independent of primary users, these results are based on the assumption that the density and traffic of primary users are very low. In fact, the expected latency $E(T_d)$ is a *decreasing function* of the density and traffic of primary users (see proof of Lemma 1, the expected first hitting time c_1 and c_2 decreases as the density and traffic of primary users increase).

IV. SIMULATION RESULTS

In this section, we carry out several sets of simulations to validate our theoretical results. Particularly, Fig. 4 and Fig. 5 provide some simulation results to support our theoretical analysis in this paper. We here consider a finite network with $n = 16$ mobile secondary users, $m = 2$ static primary users and density $\lambda = 4$ (per m^2). The time is partitioned into slots, each with length $1s$. At each time slot, the locations of each secondary user are *i.i.d* and uniformly distributed around their home points in all the three mobility. Specifically, the stationary distribution Ψ of secondary users around their home points is *Uniform* with mobility capability $\alpha = 0.8$ (m). The transmission range r of secondary users and interference range of primary users R_I are set to $r = 0.1$ meter and $R_I = 0.3$ meters respectively. And the channel usage efficiency by primary users is set 25% (i.e., on average, a primary user use the channel 1 time slot in every 4 time slots). Fig. 4 shows the complementary distribution (CCDF) of the first hitting time $\mathbb{P}(T_h(v_i, v_j) > t)$ and Fig. 5 shows CCDF of the dissemination

latency T_d on a log-log scale, under FHP and PRHP mobility. Note that the first hitting time $T_h(v_i, v_j)$ and dissemination latency T_d are *light-tailed* (exponential or gamma) under *FHP* and *PRHP* mobility, which agree with our theoretical results. Furthermore, we also find that the first hitting time $T_h(v_i, v_j)$, and thus the dissemination latency T_d of *FHP* is smaller than those of *PRHP*. This implies that *spatial inhomogeneity* of secondary users may positively influence the dissemination latency.

V. CONCLUSIONS

In this paper we have studied how fast information disseminates in cognitive radio networks under general mobility. We propose a general framework to model the node mobility, which captures most features of the existing mobility. Based on this framework, we find that there exists a critical value on *mobility capability* of secondary users, below which the latency distribution is *heavy-tailed* and above which the right tail of its distributions is bounded by some *gamma* random variable. Our theoretical results provide guidelines on mobility modeling, performance analysis and protocol design in cognitive radio networks.

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